ARALSYNC: LEARNING RECOVERY ANALYTICS USING OMR DIAGNOSTICS AND AI-DRIVEN READING ASSESSMENT AT GEN. EMILIO AGUINALDO INTEGRATED SCHOOL

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ABSTRACT

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Chapter 1

THE PROBLEM AND ITS BACKGROUND

Introduction

Education continues to evolve through the use of technology in assessing, monitoring, and intervening with learners for more efficient and effective learning, especially those who experience academic difficulties. Likewise, the use of Artificial Intelligence (AI) in education has improved assessment practices with learner profiling, performance analysis, and intervention planning (UNESCO, 2021).

In response to the learning loss of Filipino learners, the Department of Education (DepEd) launched the Academic Recovery and Accessible Learning (ARAL) Program under Republic Act No. 12028 with an aim to increase learners foundational competencies in Reading, Numeracy, and Science through diagnostic assessment, learner profiling, targeted intervention, and progress monitoring as the key components (Department of Education, 2024). As a reading assessment tool, the Philippine Informal Reading Inventory (Phil-IRI) aids teachers in determining learners oral reading profiles and identifying learners who may need reading interventions.

However, most of the public schools have been manually checking the diagnostic assessments, conducting paper-based monitoring, and analyzing

learners' oral reading performance. These processes are time-consuming and laborious, preventing teachers from learning learners' competency gap, organizing learners profiles, learning progress, and providing timely and learner-needed interventions. Consequently, learning recovery strategies may be generic for students with similar learning gaps.

When implementing the ARAL Program at Gen. Emilio Aguinaldo Integrated School, the teachers encounter the same problem in checking diagnostic assessments, evaluating learners oral reading performance, managing learner profiles, assigning interventions, and monitoring learners intervention progress. Such management tasks can be overwhelming when dealing with multiple learners, limiting the teachers time and energy for teaching and learning or intervening activities.

In response to these concerns, this study will propose AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Driven Reading Assessment at Gen. Emilio Aguinaldo Integrated School as a learning recovery analytics system for ARAL Teachers, ARAL Program Beneficiaries from Grades 7 to 10 (50 learners per grade level), and the ARAL Coordinator. The system will include automated diagnostic assessment processing with OpenCV-based Optical Mark Recognition (OMR) and learners oral reading assessments with Whisper for speech-to-text transcription and Librosa for audio feature extraction. The reading assessment results will be analyzed against the identified Phil-IRI reading criteria to determine the learner's oral reading profile. The K-Means Clustering algorithm

will group learners with similar assessment profiles, whereas the Threshold-Based Risk Classification will combine the diagnostic and reading assessment results to determine the learner's overall intervention priority. Lastly, the system will recommend intervention materials (e.g., videos, learning materials, quizzes, and activities) based on learners' competency needs.

With automated assessment processing, learner analysis, personalized intervention recommendations, and progress monitoring, AralSync will enable an efficient, organized, and data-informed implementation of the ARAL Program while saving teachers administrative and managerial time and energy and enhancing the quality of learning recovery interventions.

This study will contribute to meeting Sustainable Development Goal (SDG) 4: Quality Education by providing inclusive, equitable, and technology-driven learning opportunities for learners in need of academic interventions. It will also address SDG 9: Industry, Innovation, and Infrastructure through developing an innovative digital solution to improve assessment practices, learner monitoring, and intervention management in public schools.

Background of the Study

The Philippine basic education system continues to address learning gaps through Republic Act No. 12028, or the Academic Recovery and Accessible Learning (ARAL) Program Act. The program aims to help learners improve their foundational competencies in Reading, Numeracy, and Science through

assessment, targeted instruction, and learning recovery interventions. Teachers play an important role in identifying learners' academic difficulties, assessing their performance, and providing appropriate learning support. In reading assessment, the Philippine Informal Reading Inventory (Phil-IRI) provides procedures for evaluating learners' reading performance and identifying areas that require further support.

However, conducting diagnostic assessments, checking answer sheets, evaluating oral reading performance, and organizing learner assessment results can require considerable time and effort from teachers. Manual processing may make it more difficult to identify specific competency gaps promptly and use assessment results to plan appropriate interventions. These challenges highlight the need for a system that can organize assessment information, assist in evaluating reading performance, and support the monitoring of learners' progress.

To address these needs, this study proposes AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Driven Reading Assessment at Gen. Emilio Aguinaldo Integrated School. AralSync is a web-based system designed to support diagnostic assessment, reading fluency evaluation, learner profiling, and learning recovery planning. The study covers 200 learners, with 50 learners per grade level, and focuses on Reading, Numeracy, and Science. Its Optical Mark Recognition (OMR) module supports the processing of printed diagnostic answer sheets by detecting marked responses and comparing them

with the corresponding answer key. For oral reading assessment, the system uses speech-to-text processing and audio-feature analysis to support the evaluation of learners' reading performance. The Reading Fluency module also provides tools for reviewing reading miscues and presenting available fluency measures, including Words Correct Per Minute (WCPM), reading accuracy, pauses, and mastery classifications.

In addition, AralSync provides assessment and analytics features to help identify learners' competency gaps and areas for improvement. K-Means Clustering can group learners with similar assessment profiles, while threshold-based classification helps identify intervention priorities based on predefined rules. The system also includes modules for Comprehension Check, Skill Gap analysis, Learning Recovery, Progress Monitoring, Interventions, Recommendations, and Reports and PDF generation. These features support the organization of assessment results and help teachers identify appropriate learning resources and activities based on learners' identified needs.

Furthermore, AralSync supports the recording and review of assessment results to help teachers monitor learners' performance over time. The system aims to provide organized information that teachers and authorized school personnel can use when reviewing assessment outcomes, planning learning recovery activities, and evaluating learner progress. Through the integration of OMR processing, oral reading assessment, analytics, intervention support, and reporting, AralSync seeks to reduce the time required for assessment-related

tasks and help teachers make more informed decisions in addressing learning gaps..

Project Context

This study proposes AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Driven Reading Assessment System for Gen. Emilio Aguinaldo Integrated School, a teacher-centered web application designed to support the implementation of the ARAL Program through automated assessment processing, learner profiling, intervention recommendation, and learning recovery monitoring. The system is intended for ARAL Teachers, ARAL Program Beneficiaries from Grades 7 to 10, and the ARAL Coordinator, with 50 learners per grade level or a total of 200 learners, covering Reading, Numeracy, and Science.

The proposed system utilizes the MERN Stack (MongoDB, Express.js, React.js, and Node.js) with a Python microservice for AI and analytics processing. It aims to streamline teachers' assessment and monitoring tasks by providing a centralized platform for managing learner records, assessment results, competency deficiencies, intervention materials, and learning recovery progress.

The system will include an OpenCV-powered Optical Mark Recognition (OMR) processor to process researcher-prepared printed diagnostic assessments. The OMR component will compare learners' responses with the

answer key provided by the teacher and will be designed to achieve a 98% target recognition accuracy. Unclear, multiple, or unreadable responses will be flagged for teacher verification to prevent inaccurate assessment results.

For oral reading assessment, the system will use Whisper as a pre-trained speech-to-text model to transcribe learners' oral reading recordings, while Librosa will be used to extract audio-based features. The assessment will consider reading duration, pauses, silence duration, pacing, word-reading accuracy through Word Error Rate (WER), and Words Correct Per Minute (WCPM) as reading performance measures. The resulting reading assessment data will be analyzed using the applicable Phil-IRI criteria to classify learners into the Independent, Instructional, or Frustration reading profiles.

The system will also implement K-Means Clustering to group learners based on similarities in their diagnostic and reading assessment results. Threshold-Based Risk Classification will integrate both diagnostic and reading assessment results to determine the learner's overall intervention priority. The identified competency deficiencies, skill gaps, learner profiles, and intervention priorities will serve as the basis for recommending appropriate intervention materials.

The system will include an intervention recommendation module that recommends remediation videos, modules, learning resources, quizzes, and activities based on the identified learner deficiencies and assessment results. Teachers will be able to review and validate the recommended interventions

before they are implemented. ARAL Program Beneficiaries will be able to access assigned intervention materials, complete remediation activities, record and upload oral reading performances, and monitor their learning recovery progress through the portal. If an oral reading recording is flagged because of poor quality or incomplete content, the teacher can review the recording and request the learner to re-record it when necessary.

Teachers will be able to view learner profiles, OMR assessment results, oral reading assessment results, competency deficiencies, skill gaps, learner groupings, intervention priorities, recommended interventions, and progress monitoring data. The system will also provide analytics and PDF reports to support assessment and intervention management. The ARAL Coordinator will be able to monitor consolidated learner performance, intervention records, and overall learning recovery progress.

The system will support the measurement of learning recovery through pre-test and post-test assessments, allowing teachers to compare learners' performance before and after the implementation of interventions. Overall, AralSync aims to streamline the implementation of learning recovery interventions by automating assessment processing, analyzing learner performance, identifying competency deficiencies, grouping learners with similar assessment profiles, prioritizing intervention needs, recommending appropriate learning materials, and monitoring progress.

Purpose and Description

The purpose of this study is to develop AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Driven Reading Assessment to address the difficulties experienced by teachers in implementing learning recovery interventions through the ARAL Program at Gen. Emilio Aguinaldo Integrated School. Currently, teachers conduct manual checking, conventional assessment analysis, and paper-based learner tracking to identify learner deficiencies and monitor intervention progress. These conventional processes may result in delayed assessment analysis, generalized interventions, difficulty identifying specific learner weaknesses, and increased teacher workload when managing multiple learners.

The proposed study will develop a teacher-centered web application using the MERN Stack (MongoDB, Express.js, React.js, and Node.js) with a Python microservice to provide a learning recovery analytics system that supports assessment processing, learner profiling, intervention recommendation, and learning recovery monitoring within the ARAL Program.

In processing assessment results, the system will utilize OpenCV-powered Optical Mark Recognition (OMR) to automate the processing of researcher-prepared printed diagnostic assessments in Numeracy and Science. The system will detect and interpret learner responses and compare them with the correct answers provided by the teacher. The OMR component will have a target recognition accuracy of at least 98%, while multiple, unclear, or unreadable

responses will be flagged for teacher verification. The processed results will be analyzed to identify specific learner deficiencies and competency areas requiring intervention.

The system will also include an AI-assisted oral reading assessment feature using Whisper and Librosa. Whisper, a pre-trained speech-to-text model, will transcribe learners' oral reading recordings. The generated transcription will be compared with the reference passage to calculate Word Error Rate (WER) as a measure of word-reading accuracy, while Words Correct Per Minute (WCPM) will be used to measure reading rate. Librosa will extract audio-based reading measures such as reading duration, pauses, silence duration, and pacing. These reading performance measures will be interpreted using the applicable Phil-IRI criteria to classify learners into the Independent, Instructional, or Frustration reading profiles. Recordings that are unclear, incomplete, or unsuitable for reliable processing will be flagged for teacher review and may require the learner to submit a new recording.

A key feature of the proposed system is its learner analytics and profiling capability. The system will use K-Means Clustering to group learners according to similarities in their diagnostic and oral reading assessment results. Threshold-Based Risk Classification will then integrate the diagnostic and reading assessment results using predefined criteria to determine each learner's overall intervention priority. The identified competency deficiencies, reading

performance, learner groupings, and intervention priorities will provide the basis for recommending appropriate learning recovery interventions.

The intervention recommendation and learning recovery monitoring features will generate recommended remediation videos, modules, exercises, quizzes, and activities based on identified learner deficiencies and assessment results. ARAL Program Beneficiaries will be able to access assigned intervention materials, complete remediation activities, submit oral reading recordings, and monitor their learning recovery progress. Teachers will be able to review learner assessment results, intervention recommendations, and progress through dashboards and reports, while the ARAL Coordinator will have access to consolidated learning recovery analytics and monitoring reports. Teacher review and validation will be maintained before system-generated assessment results and intervention recommendations are used to support learning recovery decisions.

Conceptual Framework

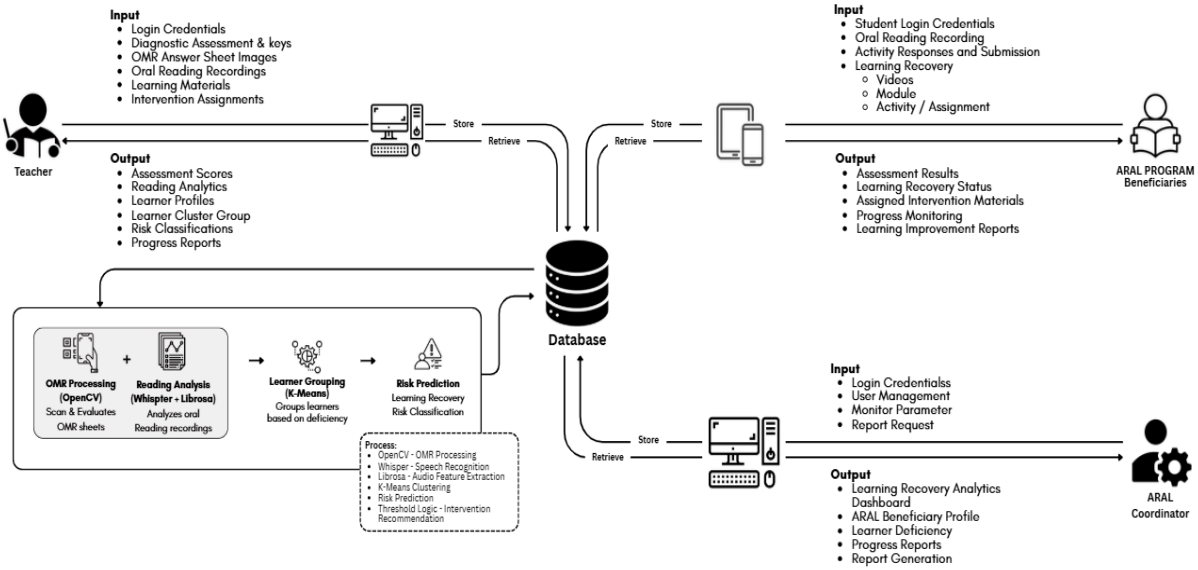


Figure 1.1. Conceptual Framework of Proposed System

Figure 1.1 illustrates the overall system process flow of AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Driven Reading Assessment at Gen. Emilio Aguinaldo Integrated School. The diagram presents the interaction among the three primary users of the system, namely the ARAL Teachers, ARAL Program Beneficiaries, and ARAL Coordinator, together with the centralized database and the system's assessment and learning analytics components.

The process begins with the ARAL Teachers uploading diagnostic assessment sheets, answer keys, intervention materials, and other necessary assessment resources into the system. ARAL Program Beneficiaries interact with the platform by accessing assigned intervention materials, completing remediation activities, submitting required outputs, and uploading oral reading recordings. Learner records, assessment results, intervention data, and oral reading recordings are stored and managed through the centralized database.

The system processes the collected assessment data through its integrated assessment and analytics modules. OpenCV-powered Optical Mark Recognition (OMR) processes the researcher-prepared diagnostic assessment sheets and determines the learners' responses based on the detected marks. Meanwhile, Whisper processes oral reading recordings through speech-to-text transcription, while Librosa extracts audio-based metrics such as reading

duration, pauses, silence, and pacing. The resulting assessment data are used to determine relevant learner performance measures.

For reading assessment, the system applies the established Phil-IRI reading criteria to determine the learner's reading profile as Independent, Instructional, or Frustration based on the applicable reading performance measures. K-Means Clustering is then used to group learners with similar assessment characteristics and learning needs. In addition, Threshold-Based Risk Classification integrates the diagnostic and reading assessment results to determine each learner's overall intervention priority. Based on these analytics, the system generates appropriate intervention recommendations, such as remediation videos, learning modules, quizzes, and activities corresponding to the identified competency deficiencies.

The processed assessment results, learner profiles, intervention priorities, recommendations, and progress information are presented through dashboards accessible to the authorized ARAL Teachers and ARAL Coordinator. Teachers can review and validate assessment results and intervention recommendations, monitor learner progress, and manage assigned interventions. The ARAL Coordinator can monitor consolidated learner performance, intervention status, and learning recovery analytics to support the overall implementation of the ARAL Program.

Objectives of the Study

This study aims to develop and evaluate AralSync, a learning recovery analytics system that automates learner assessment, competency gap identification, learner grouping, learning recovery recommendation, and progress monitoring through OMR-based diagnostics and AI-driven reading assessment at Gen. Emilio Aguinaldo Integrated School. The study is achieved through the following specific objectives that define the system's functionalities and evaluation requirements.

1. To develop an OMR-based diagnostic assessment module that processes and scores learner responses in Reading, Numeracy, and Science using teacher-provided answer keys.
2. To develop an AI-driven reading assessment that analyzes oral reading performance using speech recognition and audio feature extraction techniques.
3. To design a learner profiling and grouping identifies competency gaps and classifies learners according to their learning recovery needs.
4. To develop a learning recovery recommendation that suggests appropriate learning resources and intervention activities based on identified learner deficiencies.
5. To develop a progress monitoring and reporting to track learner performance and generate analytical reports for teachers and ARAL coordinators.

6. To evaluate the proposed system based on the ISO/IEC 25010:2011 software quality model in terms of:

6.1 Functional Suitability

6.2 Performance Efficiency

6.3 Usability

6.4 Reliability

6.5 Security

6.6 Maintainability

6.7 Portability

6.8 Compatibility

Significance of the Study

This study aims to improve the efficiency, accuracy, and effectiveness of learning recovery interventions under the Academic Recovery and Accessible Learning (ARAL) Program through automated diagnostic assessment, oral reading assessment, learner profiling, intervention recommendation, and progress monitoring. The proposed system integrates OMR technology, Artificial Intelligence (AI), and learning analytics to assist teachers in identifying learners' competency deficiencies, assessing reading performance based on the Philippine Informal Reading Inventory (Phil-IRI), recommending appropriate learning interventions, and supporting data-driven decision-making at Gen. Emilio Aguinaldo Integrated School. The following groups stand to benefit from this study:

Learners / ARAL Beneficiaries. The system will help learners identify their strengths and weaknesses in Reading, Numeracy, and Science through automated assessments and learning recovery analytics. It will provide timely learning recovery resources and interventions that can support the improvement of their foundational skills and academic performance.

Teachers. The system will assist teachers in administering and processing assessments through OMR diagnostics and AI-driven reading assessment. It will reduce the time required for manual checking, facilitate learner profiling, identify learning deficiencies, and support data-driven intervention planning. The generated reports and analytics can help teachers monitor learner progress and evaluate the effectiveness of learning recovery activities.

School Head / Principal. The system will provide school administrators with learning recovery analytics, ARAL beneficiary reports, and learner performance summaries. These insights can support informed decision-making, resource allocation, intervention planning, and monitoring of ARAL Program implementation within the school.

ARAL Coordinator. The system will serve as a monitoring and evaluation tool that enables ARAL coordinators to track learner participation, intervention implementation, and learning recovery outcomes. It can assist in identifying priority areas that require additional academic support and intervention.

Gen. Emilio Aguinaldo Integrated School. The implementation of the system may contribute to the school's efforts in improving learner achievement and strengthening the implementation of the ARAL Program through the use of technology-driven assessment and analytics. The system can provide a centralized platform for assessment management, learner monitoring, and progress evaluation.

Future Researchers. The study may serve as a reference for future researchers interested in educational analytics, learning recovery systems, artificial intelligence in education, OMR-based assessment systems, speech recognition technologies, and learner performance monitoring. The findings may provide insights for the development of similar systems in other educational institutions.

Scope and Delimitation

This study focuses on the design, development, and evaluation of AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Driven Reading Assessment at Gen. Emilio Aguinaldo Integrated School. The system is a web-based application designed to support the implementation of the Academic Recovery and Accessible Learning (ARAL) Program through diagnostic assessment, oral reading fluency evaluation, learner profiling, intervention recommendations, and progress monitoring. The study covers Reading, Numeracy, and Science and involves 200 learners, with 50 learners per grade level from Grades 7 to 10. The intended users include ARAL Teachers, the

ARAL Coordinator, and authorized ARAL Program Beneficiaries. The system provides modules for OMR Assessments, Reading Fluency, Comprehension Check, Skill Gap Analysis, Learning Recovery, Progress Monitoring, Interventions, Recommendations, and Reports and PDF generation.

The diagnostic assessment component uses OpenCV-based Optical Mark Recognition (OMR) to process researcher-prepared printed answer sheets. The system detects marked responses and compares them with teacher-provided answer keys to generate assessment results and help identify possible competency deficiencies. Unclear, multiple, or unreadable marks are subject to teacher verification. The target OMR recognition accuracy is at least 98%; however, the actual accuracy will be determined through system testing.

The oral reading assessment component uses speech-to-text processing and audio-feature analysis to support the evaluation of learners' reading performance. Whisper is used to transcribe oral reading recordings, while the generated transcript is compared with a reference passage to calculate Word Error Rate (WER). The system also provides reading fluency measures, including Words Correct Per Minute (WCPM), reading accuracy, pauses, and other available audio-based measures. Teacher review and word-level miscue calibration support the checking of recognized words and reading errors. Reading results may be interpreted using applicable Phil-IRI criteria when the required assessment procedures and scoring rules are supported by the system.

Recordings that cannot be processed reliably require teacher review or resubmission.

The system also includes analytics features for identifying competency gaps and supporting learning recovery decisions. K-Means Clustering is used to group learners with similar assessment profiles based on selected assessment features. Threshold-based classification applies predefined rules to determine intervention priorities. These results support the recommendation of appropriate learning materials, remediation activities, and other interventions for identified deficiencies in Reading, Numeracy, and Science. Teachers remain responsible for reviewing the assessment results and validating recommended interventions before implementation.

The study is limited to diagnostic assessment processing, oral reading assessment, learner profiling, learning recovery recommendations, intervention management, progress monitoring, and report generation. AralSync serves as a support tool for teachers and does not replace their professional judgment, official DepEd assessment procedures, or existing official academic records. The system does not include general academic grading, attendance monitoring, or other school administrative functions outside its defined scope. Furthermore, it does not cover integration with the Learner Information System (LIS), external government databases, or third-party Learning Management Systems (LMS). Access to the system and its features is restricted according to each user's assigned role and permissions.

Definition of Terms

This section presents and defines the terms used in the study to ensure a clear and consistent understanding of their meanings within the context. The definitions are provided based on how each term is specifically applied in the study to avoid ambiguity and guide readers in interpreting the content accurately.

Academic Recovery and Accessible Learning (ARAL) Program. – A program implemented by the Department of Education (DepEd) that provides learning interventions and remediation activities to help learners recover from learning gaps in foundational subjects.

Philippine Informal Reading Inventory (Phil-IRI). A standardized reading assessment tool developed by the Department of Education (DepEd) that serves as the basis for evaluating learners' oral reading performance and determining appropriate learning interventions in this study.

Optical Mark Recognition (OMR). – A technology used to detect and process shaded answers on printed multiple-choice answer sheets through image scanning and analysis.

Artificial Intelligence (AI). – A branch of computer science that enables systems to perform tasks that normally require human intelligence, such as speech recognition, audio analysis, and automated learner assessment.

OpenCV. – An open-source computer vision library used in the study to process and analyze printed OMR answer sheets.

Whisper. – An AI-based speech recognition model used to convert oral reading recordings into text for reading assessment.

Librosa. – A Python library used for audio and speech analysis, particularly in measuring reading duration, pauses, pacing, and hesitation patterns.

K-Means Clustering. – A machine learning algorithm used to group learners according to similarities in their assessment performance and reading metrics.

Threshold-Based Risk Classification. – A rule-based approach that classifies learners according to predefined performance thresholds to identify those requiring learning recovery interventions.

Learning Recovery. – Educational interventions and strategies designed to help learners regain lost or underdeveloped learning competencies.

Learning Recovery Analytics. – The process of collecting, analyzing, and interpreting learner assessment and intervention data to support decision-making and monitor learning recovery progress under the ARAL Program.

Diagnostic Assessment. – A type of assessment used to identify learners' strengths, weaknesses, and specific learning gaps before intervention.

Competency Deficiency. – A lack or weakness in a specific skill or learning competency demonstrated by a learner.

Teacher-Centered System. – A system primarily designed to support teachers in managing assessments, monitoring learner progress, and implementing interventions efficiently.

Analytics Dashboard. – A visual interface that displays processed learner data, assessment results, learner profiles, and competency reports for monitoring and decision-making purposes.

Word Error Rate (WER). – A measurement used to determine reading accuracy by comparing spoken words with the reference text.

Intervention Recommendation. – A system-generated suggestion of remediation videos, modules, quizzes, and learning activities based on identified learner deficiencies and mastery levels.

Learner Profiling. – The process of collecting, analyzing, and organizing learner assessment data to identify performance levels, competency deficiencies, and intervention needs.

ARAL Program Beneficiary. – A learner identified to participate in the Academic Recovery and Accessible Learning (ARAL) Program and receive

learning recovery interventions based on assessment results and identified competency deficiencies.

Mastery Level. – The degree of learner proficiency in a specific competency or learning area as determined through diagnostic assessments and performance analytics.

Reading Fluency. – The ability of a learner to read text accurately, smoothly, and with appropriate pacing and expression.

Chapter 2

REVIEW OF RELATED LITERATURES AND STUDIES

This chapter presents a comprehensive review of literature and studies from both local and foreign sources. The concepts and technologies discussed herein provide the theoretical foundation for AralSync, particularly focusing on educational interventions, computer vision for assessment, and self-hosted machine learning in education.

Local Literature

The Academic Recovery and Accessible Learning (ARAL) Program was established through Republic Act No. 12028 to address learning gaps among Filipino learners through targeted academic support and learning recovery interventions. The Department of Education's implementing guidelines for ARAL-Reading in Key Stages 2 and 3 cover learners from Grades 4 to 10 and emphasize assessment, grouping according to actual reading ability, targeted instruction, and regular progress monitoring. These guidelines identify the Philippine Informal Reading Inventory (Phil-IRI) as one of the assessment tools for determining learners who require reading support. This provides a relevant policy and instructional foundation for AralSync, which focuses on assessment,

learner profiling, intervention recommendation, and progress monitoring for Grades 7 to 10 learners (Department of Education, 2025).

The Philippine Informal Reading Inventory (Phil-IRI) is used to assess learners' reading performance and identify their reading profiles. It evaluates reading skills in English and Filipino and provides information that can help teachers determine learners' reading needs and appropriate interventions. The assessment considers reading performance and comprehension, which support the identification of learners who may require additional reading assistance (Department of Education–Regional Office XIII, 2024). These guidelines provide a basis for AralSync's oral reading assessment and learner profiling components, particularly in interpreting reading accuracy and comprehension results.

The need for more efficient assessment and monitoring is also supported by evidence regarding teacher workload in Philippine public schools. The Second Congressional Commission on Education (EDCOM II) reported that teachers spend substantial time on ancillary work, including assessment and curriculum-related tasks. The report noted that teachers spend approximately 17.8 hours per week on ancillary work, with assessment activities accounting for a considerable portion of these responsibilities. These findings highlight the importance of systems that can reduce repetitive assessment and record-management tasks and allow teachers to devote more time to instructional and intervention activities (Second Congressional Commission on Education, 2025).

Local research also demonstrates the potential of Optical Mark Recognition (OMR) for improving educational assessment processes. Luciano (2025) developed and evaluated an OMR-based software application for test item analysis, demonstrating the potential of automated recognition to reduce the time and effort associated with manual assessment processing. This supports the use of OpenCV-powered OMR in AralSync for processing uploaded diagnostic assessment sheets and generating assessment results more efficiently (Luciano, 2025).

The use of Artificial Intelligence (AI) in Philippine education is also increasing, although its implementation remains uneven. Co (2025), through a narrative review of 43 studies on AI in Philippine education, found that AI applications include teaching tools, student support systems, administrative functions, and research applications. However, the review also identified challenges involving infrastructure limitations, technical expertise, ethical concerns, and unequal access to technology, particularly in K–12 settings. These findings support the development of practical and resource-conscious AI applications such as AralSync, where AI-assisted reading assessment and learning analytics are intended to support teachers rather than replace their professional judgment.

Data privacy is another important consideration when developing educational systems for learners. The National Privacy Commission's Advisory No. 2024-03 provides guidelines on child-oriented transparency and emphasizes

appropriate protection when processing children's personal data. Since AralSync processes learner profiles, assessment results, and oral reading recordings, privacy and security measures must be considered throughout the system's design and implementation. Controlled processing of AI-assisted assessment data can help minimize unnecessary transmission of learners' information to external services (National Privacy Commission, 2024).

Overall, the reviewed local literature establishes the importance of structured learning recovery, reading assessment, automated assessment processing, teacher workload reduction, responsible AI implementation, and protection of learners' personal data. However, the reviewed sources primarily address these areas separately. This presents an opportunity for an integrated system that combines OMR-based diagnostic processing, AI-assisted oral reading assessment, Phil-IRI-based reading profiling, learner analytics, intervention prioritization, and progress monitoring within the ARAL Program. AralSync addresses this opportunity by integrating these functions into a single platform intended to support ARAL Teachers and the ARAL Coordinator in managing learning recovery interventions for Grades 7 to 10.

Foreign Literature

Transfer learning and pretrained speech models provide an effective approach for applications where collecting large amounts of task-specific training data is difficult. Radford et al. (2023) introduced Whisper, a speech recognition model trained using 680,000 hours of multilingual and multitask audio data. Their study

demonstrated that Whisper can perform competitively on speech-recognition benchmarks without requiring dataset-specific fine-tuning, making pretrained speech models suitable for applications where extensive local training data may not be available.

Whisper has also demonstrated potential for automated oral reading assessment. Henkel et al. (2023) investigated the use of large-scale speech models for evaluating oral reading fluency among students in Ghana. Their study found that Whisper V2 generated transcriptions that could be used to produce automated oral reading fluency scores that closely aligned with scores from expert human graders. This is particularly relevant to AralSync because it demonstrates the potential of speech recognition technology to support oral reading assessment in a lower-resource educational setting.

However, speech recognition accuracy can vary depending on accents, speaker characteristics, and recording conditions. Koenecke et al. (2024) evaluated Whisper across diverse English accents and speaker characteristics and found differences in recognition performance across accent groups and speech types. These findings highlight the importance of validating speech recognition systems using recordings representative of their intended users. For AralSync, oral reading recordings from selected learners can therefore be used to evaluate the reliability of the system's transcription and reading assessment process.

Whisper's adaptability can also be improved using limited examples. Wang et al. (2024) examined speech-based in-context learning for Whisper and reported that test-time adaptation using a small number of labeled speech samples could reduce Word Error Rates, including in experiments involving dialectal variation. This supports the consideration of validation and adaptation when deploying speech recognition models in environments with linguistic or pronunciation differences.

For automated assessment processing, computer vision techniques can be used to transform and analyze images before extracting information from them. In AralSync, OpenCV is used to process uploaded OMR answer sheets, including image correction and preparation for detecting marked responses. This allows paper-based diagnostic assessments to be converted into digital assessment results for subsequent learner analysis.

K-Means Clustering is an unsupervised machine learning technique that groups data according to similarities among observations. Ikotun et al. (2023) reviewed K-Means and its variants and described the algorithm as widely used because of its relatively simple implementation and low computational complexity, while also noting limitations related to cluster selection, initialization, and data characteristics. In AralSync, K-Means Clustering is applied to assessment-related features to group learners with similar performance characteristics and support learner profiling.

Threshold-based classification provides an interpretable method for converting quantitative results into predefined categories using established cutoff values. Unlike clustering, which groups observations according to similarity, threshold-based classification applies predetermined rules to determine whether a learner meets a particular performance or intervention criterion. In AralSync, Threshold-Based Risk Classification integrates diagnostic and reading assessment results to determine the learner's overall intervention priority, while the learner's reading profile is determined using the established Phil-IRI reading criteria.

The combination of automated assessment, speech processing, and learner analytics provides a foundation for developing integrated educational assessment systems. AralSync applies these technologies within a single platform by processing diagnostic assessments through OMR, analyzing oral reading recordings through speech and audio processing, grouping learners using K-Means Clustering, and applying threshold-based rules to determine intervention priority. The resulting learner profiles and assessment results are then used to support intervention recommendations and learning recovery monitoring.

Local Studies

Local studies have demonstrated the importance of systematic reading assessment to identify Filipino learners who need additional academic support. Pao (2024) conducted an analysis of oral and silent reading proficiency of Grade

7 learners and its correlation with their academic performance in English. The study used reading assessment results as a basis for an enhanced reading remediation program, demonstrating the contribution of assessment to the identification of learners in need of additional reading support. Dignos and Mangindra (2025) investigated the effect of oral and silent reading on the reading comprehension of Grade 7 learners within the MATATAG Curriculum. Their findings offer a local basis for AralSync's component on reading assessment for Grades 7 to 10.

The use of structured reading assessment is further supported by Balisoro et al. (2025) who studied reading literacy assessment in Philippine basic education with the use of the Philippine Informal Reading Inventory (Phil-IRI). Their study examined the application of Phil-IRI in Grade 7 to 10 learners, demonstrating the relevance of a structured reading assessment in the identification of learner differences and profile-based reading interventions. Fabillaran (2024) examined the reading profiles of junior high school learners based on fluency, vocabulary, and comprehension. These findings support AralSync's component on learner profiling by demonstrating the contribution of the examination of various aspects of learner reading performance to the identification of learners' needs.

The identification of learners with reading difficulties is closely related to the implementation of learning recovery interventions. Magclayo-Pante (2026) investigated the School Learning Recovery Continuity Plan (LRCP) among

Grade 10 struggling readers and found that the intervention has contributed to the improvement of learners' reading proficiency. Dionglay et al. (2023) studied an eight-week learning recovery curriculum and reported improvements in the reading performance of the participating pupils. The importance of intervention for learners with learning gaps and the role of assessment in learning recovery efforts are highlighted by these studies in relevance to AralSync's objective of utilizing assessment-based information for the identification and prioritization of learners in need of intervention.

Technology-based assessment can increase efficiency in the educational process. Luciano (2025) created and evaluated an Optical Mark Recognition (OMR)-based software application for test item analysis, demonstrating the potential of automated recognition to improve assessment processing. This finding supports AralSync's utilization of OpenCV-powered OMR to process uploaded diagnostic assessment sheets and generate assessment results with reduced manual processing.

The assessment results generated can be used in continuous monitoring of learner performance. Secreto et al. (2025) studied data-driven decision-making through real-time student progress monitoring, emphasizing the support of learner performance data for educational decision-making. This finding provides a local research basis for AralSync's progress-monitoring component, where assessment results and learner profiles can be organized to allow ARAL

Teachers and the ARAL Coordinator to monitor learner performance and make decisions about interventions.

Overall, local studies have demonstrated the relevance of systematic reading assessment, learner profiling, learning recovery interventions, automated assessment processing, and progress monitoring in Philippine educational settings. These findings provide a local research basis for the development of AralSync, focusing specifically on the component of OMR-based diagnostic assessment, AI-assisted oral reading assessment, learner profiling, intervention prioritization, and progress monitoring for Grades 7 to 10 learners.

Foreign Studies

Automated assessment technologies have been employed to enhance educational assessment efficiency, and several foreign studies have provided relevant insight into this area of research. Hernández-Mier et al. (2025) developed an unsupervised Optical Mark Recognition (OMR) approach to automatically process printed multiple-choice answer sheets. Their study demonstrated the feasibility of employing computer vision techniques to recognize marked responses and automate answer-sheet processing. This relates directly to the diagnostic assessment component of AralSync, since both involve OMR to automatically collect and process learners' responses.

Speech recognition has also been employed to support automated reading assessment. Molenaar et al. (2023) investigated automatic assessment of oral reading accuracy, employing automatic speech recognition systems such

as Whisper. Their study demonstrated the potential of speech recognition to automatically evaluate oral reading performance for reading diagnostics. This supports the use of speech recognition by AralSync for AI-assisted reading assessment, though their study focused specifically on speech assessment rather than integrating reading assessment with diagnostic testing and learner analytics.

Machine learning can further be used to identify and group learners based on their characteristics or performance. Pansri et al. (2024) integrated self-regulated learning with K-Means clustering to analyze student learning behavior. Their study demonstrated how the use of clustering could identify separate learner groups based on learning behaviors. This supports the use of K-Means by AralSync for learner profiling, though the latter employs profiling within a broader learning recovery process based on assessment results.

Machine learning has also been used to identify students who may be in need of additional academic support. Azizah et al. (2024) investigated the prediction of at-risk students during the early stage of a blended learning course, given the limited amount of data. Their study demonstrated the potential of machine learning for the identification of students at risk at an early stage. Similarly, Embarak and Hawarna (2024) developed an AI-driven system for the early detection of at-risk students based on learner and academic characteristics. These studies support AralSync's risk-classification function, though AralSync

further integrates risk identification with automated diagnostic assessment, reading assessment, and learner profiling.

Predictive analytics can also be connected to educational interventions. Alalawi et al. (2025) evaluated a learning analytics framework that connected student performance prediction with intervention actions. Their study demonstrated how predictive information could be utilized to make decisions regarding students who needed educational assistance. This is relevant to AralSync since its learner analytics are intended not only to identify students at risk but also make recommendations regarding interventions based on identified learning needs.

Learning recovery interventions provide another important basis for AralSync. Leon et al. (2026) evaluated a remote tutoring intervention designed to address learning loss among students in Peru. Their study provided evidence that targeted tutoring could support students who were experiencing learning difficulties, particularly in the context of limited resources. This supports AralSync's learning recovery objective, though the study focused on the tutoring intervention itself rather than using automated assessment and predictive analytics to determine individual learner needs.

Moreover, progress monitoring is important in determining whether interventions aid learners in improving their performance. Arafiyah et al. (2022) investigated the use of machine-learning algorithms to model and monitor learner progress, employing assessment data from Indonesian junior high school

students. Their study demonstrated the feasibility of employing machine learning to monitor learning progress. This supports AralSync's progress-monitoring component, which utilizes assessment results and learner analytics to monitor progress made by learners.

The reviewed foreign studies have demonstrated that OMR, speech recognition, machine learning, learner profiling, risk prediction, intervention, learning recovery, and progress monitoring can each contribute to different aspects of educational assessment. These studies have provided a research-based foundation for the major components of AralSync, particularly its automated OMR-based diagnostic assessment, AI-assisted oral reading assessment, learner profiling, risk classification, intervention recommendation, and progress monitoring. AralSync utilizes these established approaches to support the assessment of learner needs and the implementation of appropriate learning recovery interventions.

Synthesis of the Study

The reviewed literature and studies reveal the increasing use of technology-assisted assessment and data-driven approaches to identify learner needs and support educational interventions. Foreign studies demonstrate the applicability of Optical Mark Recognition (OMR) for automated assessment processing, speech recognition for oral reading assessment, and machine learning for learner profiling and risk identification. Other studies further demonstrate how predictive analytics and learning progress monitoring can

support intervention decisions and track changes in learner performance. These findings provide a technological foundation for the assessment and analytics processes proposed in AralSync.

The local studies provide a complementary perspective by demonstrating the relevance of these approaches within the Philippine educational context. Research involving Filipino learners emphasizes the importance of systematic reading assessment in identifying reading difficulties, while studies on learning recovery demonstrate the value of targeted interventions for learners experiencing learning gaps. Local research on OMR demonstrates the potential of automated assessment processing to improve efficiency, while studies on progress monitoring highlight the importance of using learner performance information to support educational decision-making. These findings establish the relevance of applying assessment and data-driven approaches to learning recovery within Philippine schools.

Across the reviewed sources, assessment serves as an important basis for identifying learner needs, determining appropriate support, and monitoring progress. OMR can facilitate the collection and processing of diagnostic results, speech recognition can provide automated measures for oral reading assessment, and learner analytics can organize assessment information for profiling and risk identification. Learning recovery studies further establish the importance of using assessment information to guide interventions and monitor learner development. This relationship is consistent with AralSync's intended

process of using assessment results to support learner profiling, intervention prioritization, and progress monitoring.

However, the reviewed literature and studies primarily examine these assessment, analytics, reading, and learning recovery approaches through their respective applications. The findings therefore provide separate technological and educational foundations rather than a specific implementation designed to serve the ARAL Program and the needs of Grades 7 to 10 learners. The local literature and studies also emphasize considerations relevant to Philippine schools, including the need for efficient assessment processes, appropriate learning interventions, and responsible handling of learner information.

Overall, the reviewed literature and studies provide a research-based foundation for AralSync's major functions. The findings support the use of OMR for diagnostic assessment, AI-assisted speech processing for oral reading assessment, machine learning for learner profiling and risk classification, and learner analytics for intervention prioritization and progress monitoring. Based on these findings, AralSync applies these approaches within an ARAL-focused learning recovery context to assist ARAL Teachers and the ARAL Coordinator in identifying learner needs, supporting intervention decisions, and monitoring learner progress among Grades 7 to 10

Chapter 3

METHODOLOGY AND DESIGN

This chapter presents the methodology and design used in the development of AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Driven Reading Assessment. It discusses the research design, development approach, participants, data gathering procedures, system development process, and techniques used to evaluate the proposed system. These methods provide a structured process for developing and assessing the system based on the identified needs of the ARAL Program.

Technical Backgrounds

The current process of evaluating diagnostic assessments and oral reading performance at Gen. Emilio Aguinaldo Integrated School primarily relies on manual checking, paper-based learner records, and conventional reading assessment procedures. Although these methods support the implementation of the ARAL Program, they may result in delayed feedback, increased teacher workload, difficulty identifying learner deficiencies, and challenges in organizing intervention records and monitoring learner progress.To address these concerns, the proposed study introduces a cloud-ready web application integrated with a local AI and analytics processing microservice. The system adopts a multi-tier

client-server architecture to support automated assessment processing, learner profiling, intervention recommendation, and centralized learning recovery monitoring.

The backend is developed using Next.js with React, TypeScript, and Tailwind CSS. Next.js Server Route Handlers, running in the Node.js environment, manage server-side operations such as user authentication, learner records, intervention management, and communication between system components. MongoDB is used to store system data, while Mongoose supports data modeling and validation. A separate Python microservice powered by FastAPI and Uvicorn handles computational tasks related to assessment processing and analytics.

The proposed system utilizes OpenCV to automate the processing of researcher-prepared OMR-based diagnostic assessments from uploaded answer sheets. The system detects and interprets marked responses and compares them with the correct answers provided by the teacher. The OMR component targets a recognition accuracy of at least 98%. Correct, unanswered, multiple, unclear, or unreadable responses are identified during processing, with ambiguous responses flagged for teacher verification to minimize inaccurate results. The system also supports image-based answer sheets and multi-page scanned PDF uploads, subject to the implemented processing capabilities.

For oral reading assessment, the system uses the Whisper large-v3 speech-to-text model through the Groq Cloud API to transcribe learners' oral

reading recordings. The generated transcription is compared with the reference passage to calculate Word Error Rate (WER) as a measure of transcription accuracy. The system also calculates Words Correct Per Minute (WCPM) to determine the learner's reading rate. The generated results support the teacher's review of the learner's reading performance. Meanwhile, Librosa is used to extract audio-based features, including reading duration, pauses, silence duration, and pacing. These measures provide additional indicators of learners' oral reading performance and support the assessment process in accordance with the applicable Philippine Informal Reading Inventory (Phil-IRI) criteria. Based on the assessment results and established classification criteria, learners are classified into the Independent, Instructional, or Frustration reading profiles.

Oral reading recordings must contain sufficiently clear and complete audio for processing. Recordings affected by excessive background noise, low volume, incomplete reading, or other quality issues that may affect transcription and audio analysis are flagged for teacher review. When necessary, the teacher may request a new recording before the assessment results are finalized.

The system also uses Scikit-learn's K-Means Clustering to group learners based on similarities in their diagnostic and oral reading assessment results. Meanwhile, threshold-based risk classification integrates diagnostic and reading assessment results using predefined criteria to determine each learner's overall intervention priority. The identified learning difficulties and intervention priorities serve as the basis for recommending appropriate learning recovery interventions.

The front-end interface is developed using React.js within the Next.js framework, with TypeScript and Tailwind CSS supporting development and interface styling. It provides a responsive and user-friendly platform for ARAL Teachers, ARAL Program Beneficiaries, the ARAL Program Coordinator, and the School Head or Principal. MongoDB serves as the database management system, while Mongoose supports data modeling and database operations. The database stores and manages learner profiles, assessment results, oral reading records, intervention data, and learning recovery progress.

Approaches and Techniques

This study uses a Developmental Research Design, which focuses on the systematic development and evaluation of a technological solution to address identified problems in existing practices. This design is appropriate for developing AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Driven Reading Assessment for the ARAL Program at Gen. Emilio Aguinaldo Integrated School.

During the initial phase, descriptive analysis was used to examine the existing processes for administering diagnostic assessments, evaluating oral reading performance, managing learner records, assigning interventions, and monitoring learner progress. The identified challenges and inefficiencies served as the basis for determining the functional and non-functional requirements of the proposed system.

During the development phase, the researchers designed and implemented AralSync using a web-based application integrated with a Python-based microservice for assessment processing and analytics. The system incorporates OMR-based diagnostic assessment, AI-assisted oral reading assessment, learner profiling, intervention management, and progress monitoring. OpenCV supports the processing of OMR answer sheets, while the Groq Cloud API with the Whisper large-v3 model supports speech-to-text transcription. Librosa extracts audio features, including reading duration, pauses, and silence duration, to provide additional information for reviewing oral reading performance.

The study also applies data analysis techniques to support learner profiling and intervention prioritization. Word Error Rate (WER) measures differences between the generated transcription and the reference passage, while Words Correct Per Minute (WCPM) represents the learner's reading rate. K-Means clustering is used to group learners according to selected assessment indicators. Threshold-based risk classification may also be applied using predefined criteria to identify learners who require additional learning support. The results are intended to assist teachers in reviewing learners' needs and selecting appropriate learning recovery interventions.

Finally, the developed system will undergo testing and evaluation to assess the functionality, performance, and reliability of its major features. Diagnostic assessment processing will be checked against known answer-sheet

responses, while oral reading assessment outputs will be reviewed for transcription and reading-measure accuracy. The evaluation will also examine learner profiling, intervention management, and progress monitoring. Responses or recordings that cannot be reliably processed should be referred for teacher verification. This helps ensure that system-generated results are reviewed appropriately before they are used to support learning recovery decisions.

Methodology

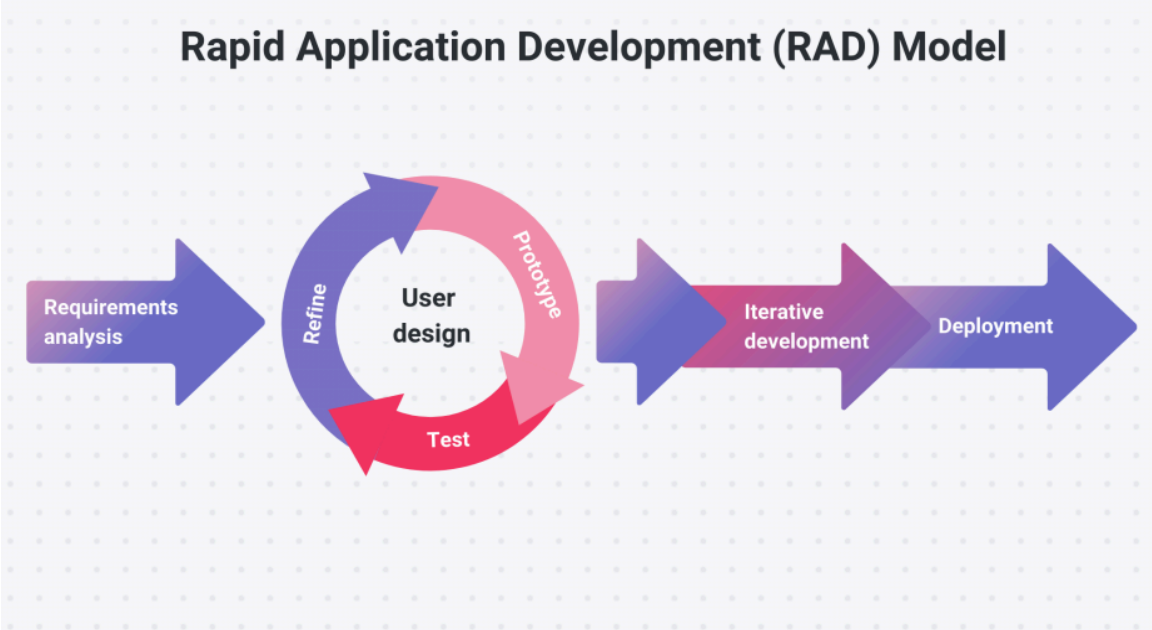


Figure 3.1 Rapid Application Development Model

Figure 3.1 illustrated the Rapid Application Development (RAD) Model utilized in developing AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Driven Reading Assessment at Gen. Emilio Aguinaldo Integrated School. The model emphasized rapid development, user feedback, iterative refinement, testing, and deployment.

The development began with Requirements Planning, where the researchers identified the requirements and challenges related to diagnostic assessment processing, oral reading assessment, learner profiling, intervention management, and learning recovery monitoring. The needs of the intended users were also considered in determining the system's functional and non-functional requirements.

The User Design phase focused on designing the system architecture, database structure, user interfaces, dashboards, and assessment workflows. The system design incorporated the Next.js web application, MongoDB database, and Python-based microservice for assessment processing and analytics. It also covered the four major components of AralSync: OMR Diagnostic Processing, AI-Assisted Reading Assessment, Learner Analytics and Profiling, and Intervention Recommendation and Monitoring.

The Iterative Development phase involved developing, integrating, testing, and refining the system components based on identified issues and feedback. The researchers evaluated the implemented features, including OMR processing, oral reading assessment, learner profiling, intervention management, and progress monitoring. The results of testing were used to identify areas requiring improvement.

Finally, the Deployment phase involved configuring the system, conducting final checks, and preparing the intended users to operate the application. The completed system was then prepared for evaluation in the ARAL Program

environment to determine whether its implemented features met the identified requirements.

Requirement Analysis

AralSync's requirements were identified based on the needs of its intended users and the objectives of the proposed learning recovery system. The system is designed to support students or ARAL Program beneficiaries, teachers, the ARAL Coordinator, and the Principal. Each user role is provided with access to functions appropriate to its responsibilities. Students can access assigned learning activities and view available assessment information, while teachers manage learner records, process diagnostic assessments, review reading assessment results, and organize learning interventions. The ARAL Coordinator and Principal can access consolidated reports and monitoring information based on their assigned permissions.

The system's functional requirements cover user authentication, learner information management, diagnostic assessment processing, oral reading assessment, learning analytics, learning resource management, intervention assignment, progress monitoring, and report generation. The Optical Mark Recognition (OMR) module processes uploaded answer-sheet images by detecting learners' marked responses and comparing them with the answer key provided by the teacher. For the supported 50-item assessment workflow, the system generates assessment results that help identify possible competency deficiencies and areas requiring further instruction. Unclear, multiple, or

unreadable marks are flagged for teacher verification to reduce the possibility of incorrect scoring.

The oral reading assessment module processes uploaded reading recordings using speech-to-text transcription and audio-feature analysis. The generated transcript is compared with the reference passage to calculate the Word Error Rate (WER), which measures the differences between the recognized words and the reference text. The system also analyzes audio characteristics, such as reading duration, pauses, and pacing, to provide additional information about the learner's oral reading performance. Words Correct Per Minute (WCPM) is used as a reading fluency measure when the required timing and word-level scoring information is available. These results help teachers review learners' reading performance and identify areas that may require additional support.

The analytics component identifies competency gaps and groups learners with similar assessment profiles using K-Means Clustering. The system also applies predefined threshold-based rules to classify learners according to their assessment results and determine intervention priorities. These analytical outputs provide teachers with information for understanding learners' academic needs and selecting appropriate learning recovery activities. The resulting classifications and recommendations serve as decision-support tools and do not replace teacher evaluation or official reading assessment procedures.

Furthermore, AralSync provides learning resource management, intervention assignment, progress monitoring, and report generation to support the implementation of learning recovery activities. Teachers can use assessment results and identified competency gaps to select appropriate learning materials and provide targeted interventions. The system generates summaries and reports that help authorized users review learner performance and monitor assessment outcomes. Role-based access control is applied to restrict system functions and learner information according to each user's permissions. Through these functions, AralSync aims to support more organized assessment processing, data-driven learner profiling, and informed intervention planning.

Functional Requirements

Table 3.1 Functional Requirements

ID	Requirement Description	Priority
FR1	The system shall allow authorized users to log in and access functions according to their assigned roles.	High
FR2	The system shall allow authorized users to view and manage learner information according to their permissions.	High
FR3	The system shall allow teachers to generate diagnostic questionnaires based on selected subjects, competencies, assessment scope, and item count.	High

FR4	The system shall allow teachers to select assessment materials from available sources, including DepEd ARAL reference materials and uploaded content.	High
FR5	The system shall generate answer keys for supported diagnostic assessments.	High
FR6	The system shall allow teachers to scan or upload learner answer sheets for OMR processing.	High
FR7	The system shall detect marked answers and compare them with the corresponding answer key.	High
FR8	The system shall present assessment results to help teachers identify learner performance and competency gaps.	High
FR9	The system shall allow teachers to conduct oral reading fluency assessments using a selected passage and a learner's recorded or captured reading.	High
FR10	The system shall provide word-level calibration tools that allow teachers to review and mark oral reading miscues.	High

FR11	The system shall calculate reading fluency measures, including WCPM and reading accuracy, based on the implemented scoring procedure.	High
FR12	The system shall track reading miscues, including substitutions, omissions, insertions, and repetitions.	High
FR13	The system shall display oral reading assessment results, including available fluency measures and mastery classifications.	High
FR14	The system shall provide a Comprehension Check module for assessing learners' reading comprehension.	High
FR15	The system shall analyze learner assessment results to identify skill gaps and areas requiring improvement.	High
FR16	The system shall group similar learner assessment profiles using K-Means Clustering.	Medium
FR17	The system shall apply predefined threshold-based rules to classify	High

	intervention priority based on assessment results.	
FR18	The system shall provide learning recovery recommendations based on identified learner needs.	High
FR19	The system shall allow authorized users to manage or assign appropriate learning interventions and resources.	High
FR20	The system shall provide progress monitoring information based on available learner assessment results.	High
FR21	The system shall generate assessment summaries and reports through the Reports & PDF module.	High
FR22	The system shall allow authorized users to review previous reading assessment results through the result history feature.	Medium

Non-Functional Requirement

Table 3.2 Non-Functional Requirement

ID	Requirements Description	Priority
NFR1	The system shall provide a user-friendly and responsive interface for ARAL Teachers, ARAL Program Beneficiaries, and the ARAL Coordinator.	High
NFR2	The system shall process uploaded OMR answer sheets accurately and efficiently.	High
NFR3	The system shall securely store learner records, assessment results, oral reading recordings, and intervention data within the database.	High
NFR4	The system shall restrict unauthorized access through authentication and role-based access control.	High
NFR5	The system shall maintain data consistency and reliability during assessment processing and analytics generation.	High
NFR6	The system shall support simultaneous access of multiple users within the school environment.	Medium
NFR7	The system shall provide fast retrieval of learner records, reports, and analytics dashboards.	Medium
NFR8	The system shall be compatible with modern desktop and web browsers.	Medium

NFR9	The system shall maintain acceptable system performance during uploading and processing of assessment sheets and audio recordings.	High
NFR10	The system shall provide maintainable and modular system architecture for future updates and enhancements.	Medium
NFR11	The system shall ensure the accuracy of assessment analytics, learner profiling, and intervention recommendations.	High
NFR12	The system shall provide backup and recovery mechanisms for learner data and system records.	Medium

Population, Sample Size, and Sampling Technique

The target population of the study consists of the stakeholders directly involved in the implementation and monitoring of learning recovery interventions under the ARAL Program at Gen. Emilio Aguinaldo Integrated School. The intended users and respondents of the proposed system are the ARAL Program Beneficiaries, ARAL Teachers, and the ARAL Coordinator.

For the learner respondents, the researchers will use stratified sampling to ensure representation from each target grade level. The ARAL Program Beneficiaries will be grouped into four strata according to grade level: Grade 7, Grade 8, Grade 9, and Grade 10. A total of 50 learners will be selected from each grade level, resulting in 200 learner respondents.

The ARAL Teachers and ARAL Coordinator will also participate as respondents because of their direct involvement in administering assessments, managing interventions, monitoring learner progress, and overseeing the implementation of the ARAL Program. Their participation will provide feedback from the instructional and administrative perspectives of the proposed system.

The selected respondents will provide feedback based on their respective roles and interactions with AralSync. ARAL Teachers will evaluate functions related to assessment processing, learner profiling, intervention recommendation, and progress monitoring. ARAL Program Beneficiaries will provide feedback on the usability and accessibility of features such as assigned intervention materials, remediation activities, oral reading submission, and progress monitoring. The ARAL Coordinator will provide feedback on monitoring, analytics, and reporting functions.

The respondents' feedback will be used to evaluate the proposed system's functional suitability, performance efficiency, usability, reliability, and security, particularly in supporting diagnostic assessment processing, oral reading assessment, learner profiling, intervention recommendation, and learning recovery monitoring.

Description of Respondents

The respondents of this study will be selected from the stakeholders directly involved in the implementation and monitoring of the ARAL Program at

Gen. Emilio Aguinaldo Integrated School. The respondents will consist of the ARAL Coordinator, ARAL Teachers, and ARAL Program Beneficiaries, representing the administrative, instructional, and learner perspectives in the evaluation of the proposed AralSync system.

The ARAL Coordinator oversees the implementation and monitoring of learning recovery interventions and reviews consolidated learner performance, intervention records, and program reports. The ARAL Coordinator's evaluation will focus on the system's functionality, usability, reliability, and capability to support learning recovery monitoring, analytics, and report generation.

The ARAL Teachers administer diagnostic assessments, manage learner records and intervention materials, review assessment results, validate system-generated reading assessment results and intervention recommendations, and monitor learner progress. Their feedback will contribute to the evaluation of the system's functional suitability, usability, performance efficiency, reliability, and security in supporting learning recovery activities.

The ARAL Program Beneficiaries will comprise learners from Grades 7 to 10, with 50 learners selected from each grade level, for a total of 200 learner respondents. The learners will use the system to access assigned intervention materials, complete remediation activities, submit oral reading recordings, and monitor their learning recovery progress. Their feedback will be used to evaluate the system's usability, accessibility, performance, and reliability from the learner's perspective.

The participation of the ARAL Coordinator, ARAL Teachers, and ARAL Program Beneficiaries will enable the evaluation of AralSync from administrative, instructional, and learner perspectives. Their feedback will provide information regarding the system's ability to support assessment processing, learner profiling, intervention management, and learning recovery monitoring, as well as its overall software quality based on the selected evaluation criteria.

Research Instrument

The primary research instrument for quantitative data collection is a standardized evaluation questionnaire based on the ISO/IEC 25010 Software Quality Model. The questionnaire is designed to evaluate the quality of the proposed AralSync system based on relevant software quality characteristics, including Functional Suitability, Performance Efficiency, Usability, Reliability, and Security.

The questionnaire uses a 5-point Likert scale to measure the respondents' level of agreement with each statement, where 5 = Strongly Agree, 4 = Agree, 3 = Neutral, 2 = Disagree, and 1 = Strongly Disagree. The questionnaire will be administered to the identified respondents to gather quantitative feedback regarding the system's functionality, efficiency, ease of use, reliability, and security.

Data Gathering Procedures

The data gathering procedure will be implemented systematically to guarantee that the required data for the study and system evaluation will be properly gathered. First, the researchers will submit a request letter to the administration of Gen. Emilio Aguinaldo Integrated School to seek permission to conduct the study and system evaluation.

Once approved, the researchers will set up a coordination session with the ARAL Coordinator and Teachers to identify the selected respondents and prepare the needed assessment materials and system data. The AralSync system will then be deployed in a testing environment. The selected ARAL Teachers, ARAL Program Beneficiaries, and ARAL Coordinator will receive an orientation regarding the system's purpose, features, and proper use before the evaluation.

During the system evaluation, the respondents will interact with the features relative to their respective roles. Teachers will evaluate the system functions related to diagnostic assessment processing, oral reading assessment, learner profiling, intervention recommendation, and learning recovery monitoring. ARAL Program Beneficiaries will access the assigned intervention materials, complete the remediation activities, upload their oral reading recordings, and monitor their progress. The ARAL Coordinator will evaluate the monitoring dashboards, reports, and consolidated learning recovery information.

After the respondents have interacted and evaluated the system, the researchers will administer the ISO/IEC 25010-based evaluation questionnaire to

assess the proposed system according to the selected software quality characteristics. The completed questionnaires will then be gathered, checked for completeness, tabulated, and subjected to the appropriate statistical treatment.

Finally, the researchers will analyze the gathered data to determine the software's quality, usability, functionality, reliability, performance efficiency, and security based on the respondents' evaluations. The results will become the basis for identifying areas for improvement and determining whether the proposed system satisfies the set requirements of its intended users.

Statistical Treatment of Data

To systematically analyze the quantitative data gathered from the ARAL Coordinators, Teachers, and ARAL Program Beneficiaries, appropriate statistical tools will be utilized.

1.Percentage. This statistical tool will be used to determine the distribution of respondents according to their respective groups and grade levels.

$P = (f / N) \times 100$

Where:

P = Percentage

f = Frequency of responses

N = Total number of respondents

2.Weighted Mean. This statistical tool will be used to determine the average rating of the respondents regarding the proposed system's Functional

Suitability, Performance Efficiency, Usability, Reliability, and Security based on the 5-point Likert scale.

$WM = (\Sigma wx) / N$

Where:

WM = Weighted Mean

w = Weight assigned to each Likert-scale response

x = Frequency of responses

N = Total number of respondents

Table 3.3. Interpretation of Mean Scores for Likert Scales

Scale	Range	Verbal Interpretation
5	4.20 - 5.00	Strongly Agree (SA)
4	3.40 - 4.19	Agree (A)
3	2.60 - 3.39	Neutral (N)
2	1.80 - 2.59	Disagree (D)
1	1.0 - 1.79	Strongly Disagree (SD)

Hardware

Table 3.4. Hardware Requirements for System Implementation

Component	Minimum Specification	Recommended Specification	Purpose

Processor (CPU)	Intel Core i3 (10th Gen) / AMD Ryzen 3 (3000 Series)	Intel Core i5 (10th Gen) or higher / AMD Ryzen 5 (3000 Series) or higher	Handles system processing, OMR image uploads, and dashboard analytics computations.
Memory (RAM)	16 GB RAM	16 GB RAM or higher	Supports multitasking, smooth web portal performance, and report generation.
Storage	256 GB SSD	512 GB SSD or higher	Stores downloaded reports, offline backups, and multimedia intervention materials.
Monitor	1366 × 768 Resolution	Full HD 1920 × 1080 Resolution	Displays learning recovery dashboards, learner profiles, and system interfaces.
Keyboard and Mouse	Standard USB Keyboard and Mouse	Wireless or ergonomic peripherals	Enables teacher and coordinator interaction with the system.

Network Device	Ethernet / Wi-Fi Adapter	High-speed Wi-Fi or Gigabit Ethernet	Supports connectivity for cloud access, AI processing, and real-time data sync.
Printer	Standard inkjet or laser printer	Continuous Ink Supply System (CISS) Color Printer	Used for printing physical OMR answer sheets and learning recovery reports.
Scanner / Camera	Smartphone camera with at least 8 MP	High-resolution smartphone camera or document scanner	Captures clear images of OMR answer sheets for OpenCV-based processing.
Audio Device	Standard Smartphone Microphone	Headset or external microphone with Noise-Canceling Microphone	Records oral reading performances for Whisper transcription and Librosa-based audio analysis. .

The hardware requirements for AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Driven Reading Assessment are established to support the web-based system, OMR processing, AI-assisted oral reading

assessment, learner analytics, and learning recovery monitoring. The system can be accessed through desktop or laptop computers for ARAL Teachers and the ARAL Coordinator, while compatible computers or mobile devices may be used by ARAL Program Beneficiaries for accessing intervention materials and submitting oral reading recordings.

The minimum hardware requirements include an Intel Core i3 10th Generation or AMD Ryzen 3 3000 Series processor, 16 GB RAM, and a 256 GB SSD. The recommended configuration includes an Intel Core i5 10th Generation or AMD Ryzen 5 3000 Series processor or higher, 16 GB RAM or higher, and a 512 GB SSD or higher. These specifications support system operation, file processing, dashboard access, report generation, and storage of assessment files, recordings, reports, and intervention materials.

For OMR processing, a smartphone camera with at least 8 MP or a document scanner will be used to capture clear images of researcher-prepared OMR diagnostic sheets. Clear and properly captured images are necessary to support reliable OpenCV-based response detection. For oral reading assessment, a smartphone or computer microphone may be used, while a headset or external noise-canceling microphone is recommended to improve recording quality for Whisper transcription and Librosa-based audio analysis.

An Ethernet or Wi-Fi connection is required to access the web application, upload assessment files and oral reading recordings, retrieve results, and synchronize system data. Printers will also be used to produce the

researcher-prepared OMR diagnostic sheets and other necessary learning recovery materials. External or backup storage may additionally be used for storing assessment records, reports, recordings, and other system-related files.

Overall, the specified hardware provides the necessary resources for assessment processing, oral reading assessment, learner analytics, intervention management, and learning recovery monitoring within the proposed AralSync system.

Software

This section describes the software environment used to develop and operate AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Assisted Reading Assessment. The selected technologies support the system's efficient processing of assessment data, automated reading diagnostics, learner profiling, and intervention recommendations. These technologies were chosen to provide a reliable, scalable, and secure platform that can effectively support learning recovery and reading intervention activities within the institutional setting.

Software Requirements

Table 3.5. Software Specifications for System Development

Category	Component	Specification

Operating System	Windows	Windows 10/11
Frontend Framework	React.js	Web Interface
Backend Runtime	Node.js & Express.js	Server Development
Database	MongoDB	Data Storage
Programming Language	Python	AI Processing
AI Tools	Whisper & Librosa	Reading Assessment
OMR Tools	OpenCV	AnswerSheet Processing
Browser	Google Chrome	System Access

The table represents the software required for the development and operation of the AralSync: Learning Recovery Analytics Using OMR Diagnostics and AI-Driven Reading Assessment. The identified software is meant to provide the necessary environment to develop the proposed solution, process assessments, conduct learner analytics, and manage the database. Windows is used as an operating system, whereas React .js, Node .js, and Express .js are applied for front-end, back-end, and database development. MongoDB is another software required for the proposed solution, which ensures the effective

management of learners' records and assessments. Additionally, Python is necessary for the proper functioning of the system as it implements artificial intelligence (AI) and analytics modules. Such software as OpenCV, Whisper, and Librosa are taken into consideration for assessment processing, reading assessment, and audio processing, respectively. Finally, Google Chrome is suggested to be used as a web browser for effective accessing and testing of the developed system. Overall, the software described above is required for efficient development and effective functioning of the proposed solution.

Peopleware

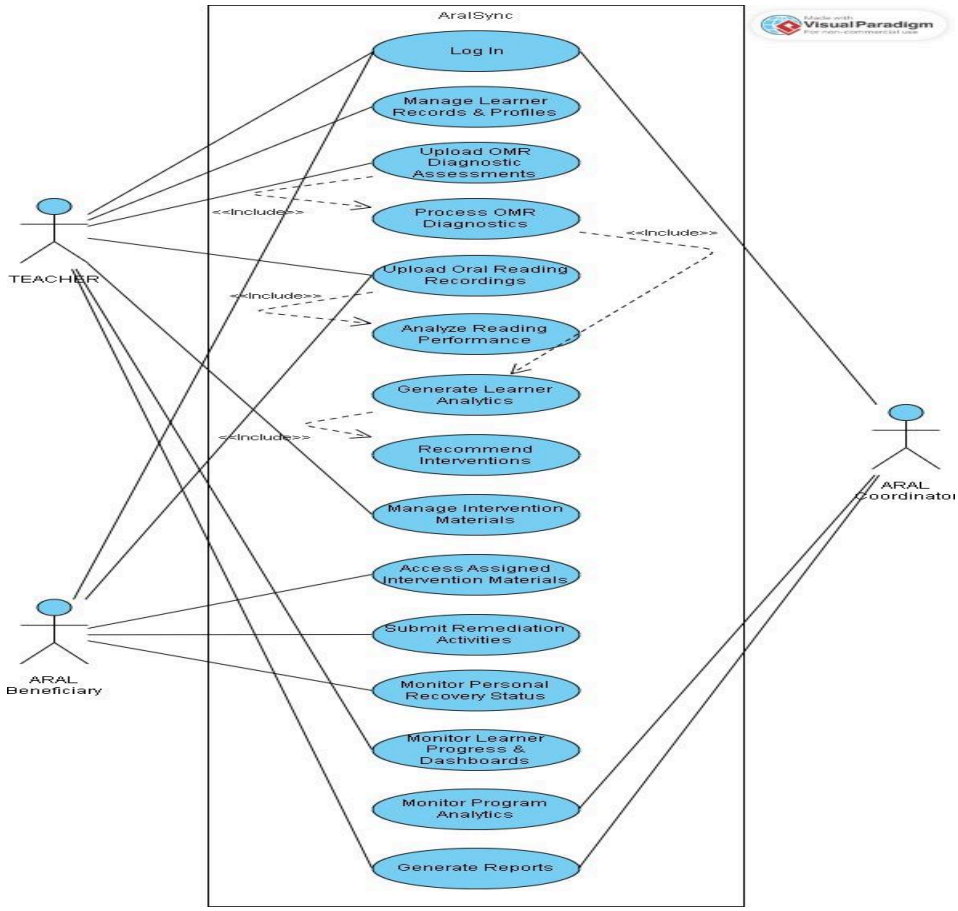


Figure 3.2 Use Case Actors and Roles

Figure 3.2 The figure illustrates the Use Case Diagram of the AralSync: Learning Recovery Analytics System, showing the interactions between the system and its three primary actors: Teacher, ARAL Beneficiary, and ARAL Coordinator.

The Teacher is responsible for managing learner records and profiles, uploading OMR diagnostic assessments and oral reading recordings, managing intervention materials, monitoring learner progress, and generating reports. The

teacher also initiates the assessment processes that enable learner analytics and intervention recommendations.

The ARAL Beneficiary interacts with the system by accessing assigned intervention materials, submitting remediation activities, uploading oral reading recordings, and monitoring their personal learning recovery status.

The ARAL Coordinator oversees the implementation of the ARAL Program by monitoring learner progress, viewing program analytics, and generating reports to support decision-making and learning recovery management.

The system processes uploaded assessment data through OMR diagnostics and reading performance analysis to generate learner analytics. Based on the identified learner deficiencies and performance levels, the system automatically recommends appropriate intervention materials and supports continuous monitoring of learner progress. Through these functions, AralSync facilitates a more efficient, organized, and data-driven implementation of learning recovery interventions under the ARAL Program.

Network

Table 3.6: Network Requirements and Configuration

Category	Requirement Specification	Technical Configuration
Network Type	Local Area Network (LAN)	School-based LAN Environment
Internet	Broadband Internet	Minimum 10 Mbps

Connection	Connection	
Network Device	Router/Modem	Supports TCP/IP Communication
Server Connectivity	Client-Server Communication	Connected through LAN or Internet
Communication Protocol	TCP/IP	Standard Network Protocol
Browser Access	Web Browser	Google Chrome (Latest Version)
Wireless Access	Wi-Fi Connectivity	IEEE 802.11n/ac Standard

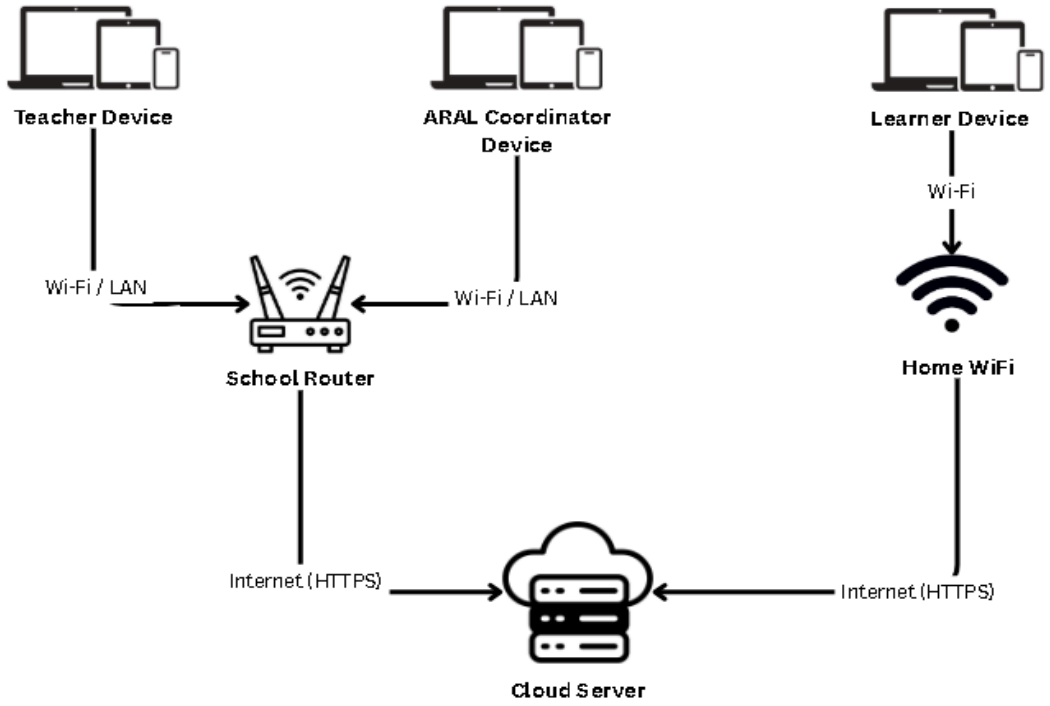


Figure 3.3 Network Diagram

Figure 3.3 presents the network requirements and configuration necessary for the operation of the AralSync system. The system utilizes a client-server architecture that requires a Local Area Network (LAN) and internet connectivity to facilitate communication between users, the web application, and the database server. Network devices such as routers and modems provide connectivity, while TCP/IP serves as the standard communication protocol. Users can access the system through a web browser, such as Google Chrome, using either wired or wireless network connections. These network components ensure reliable system accessibility, data transmission, and communication among teachers, learners, and administrators.

Dataware

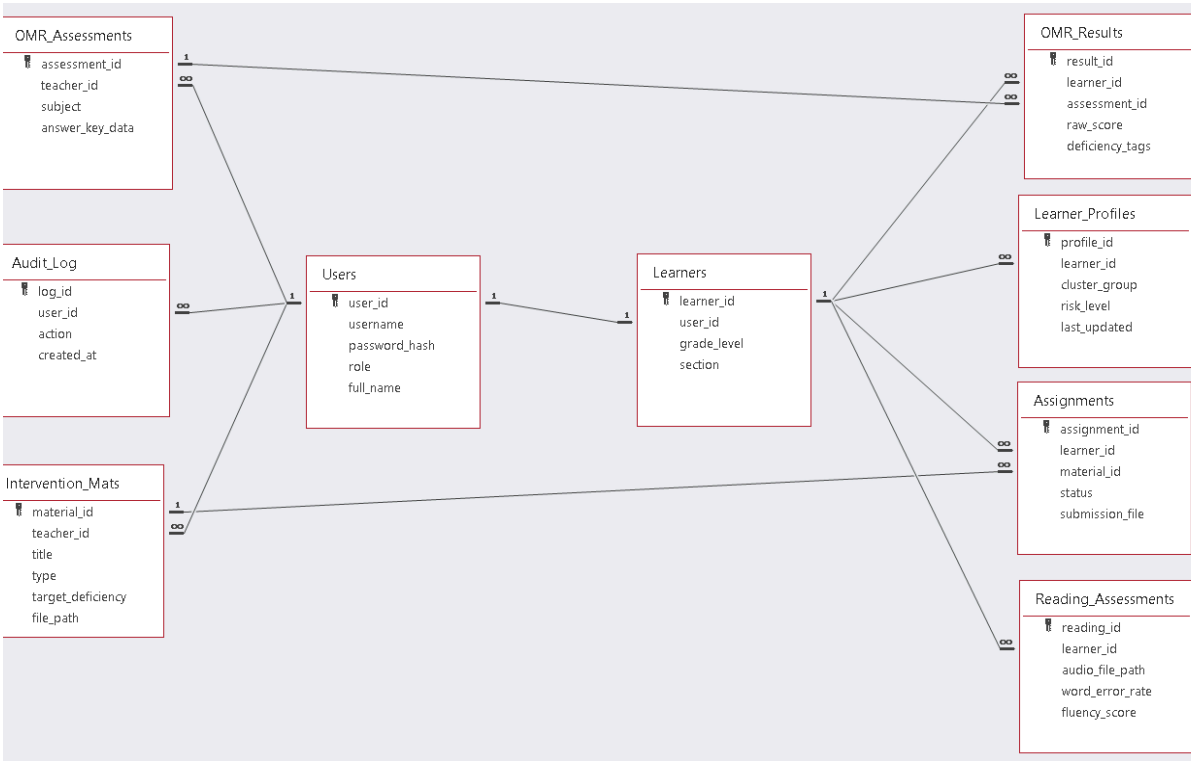


Figure 3.4: Database Schema Diagram

Figure 3.4 illustrates the Entity-Relationship Diagram (ERD) of the AralSync: Learning Recovery Analytics System, featuring a centralized database architecture composed of interconnected tables designed to support learner assessment, profiling, intervention management, and progress monitoring. At its core, the Users and Learners entities manage user accounts and learner information, serving as the foundation for assessment and analytics processes. Diagnostic assessment data are stored in OMR_Assessments and processed through OMR_Results, while oral reading performance records are maintained in Reading_Assessments. Learner performance data are consolidated within Learner_Profiles, where clustering and risk classifications are stored to support intervention planning. Learning recovery resources are managed through Intervention_Mats and distributed to learners via Assignments, enabling the tracking of remediation activities and submissions. To ensure accountability and system security, all user activities are recorded in Audit_Log, providing a complete record of system transactions and operations.

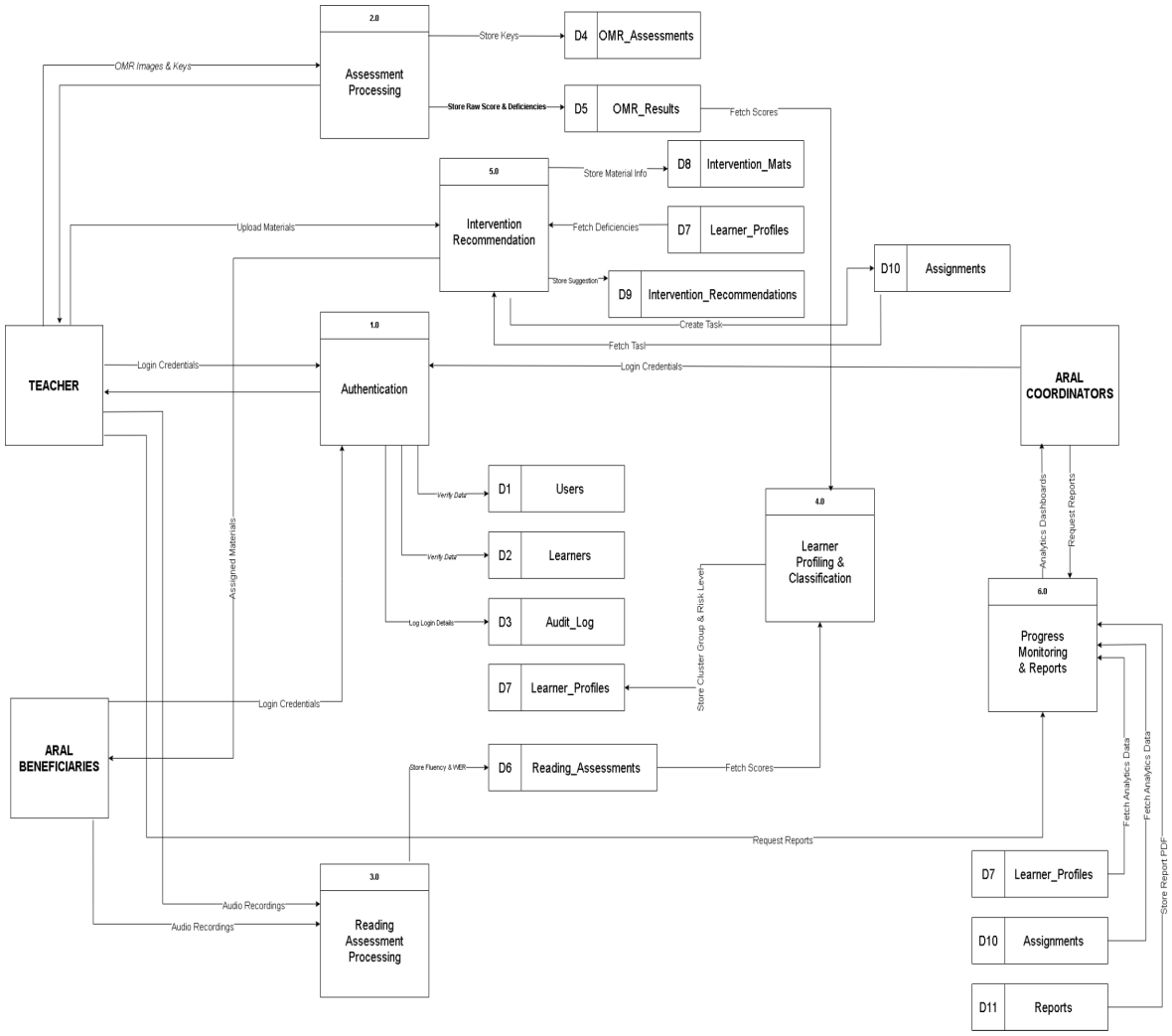


Figure 3.5: Data Flow Diagram

Figure 3.5 The figure illustrates the Data Flow Diagram (DFD) of the AralSync: Learning Recovery Analytics System, showing the flow of data among the system's primary users, processes, and data stores. The system involves three main external entities: Teachers, ARAL Beneficiaries, and ARAL Coordinators.

The process begins with Authentication, where users provide login credentials that are verified through the Users database. Teachers upload OMR answer sheet images, answer keys, and intervention materials, while ARAL Beneficiaries submit oral reading recordings and access assigned learning materials.

The uploaded assessment data are processed through the Assessment Processing module, which evaluates OMR assessments and stores learner scores and identified competency deficiencies. Oral reading recordings are handled by the Reading Assessment Processing module, which analyzes reading performance and stores fluency and accuracy results.

Assessment and reading results are then forwarded to the Learner Profiling and Classification module, where learners are grouped according to performance levels and assigned corresponding risk classifications. Based on the generated learner profiles and identified deficiencies, the Intervention Recommendation module automatically recommends appropriate learning recovery materials and creates intervention assignments for learners.

Finally, the Progress Monitoring and Reports module consolidates learner profiles, intervention records, and assessment results to generate analytics dashboards, progress reports, and learning recovery reports for teachers and ARAL coordinators. Through these interconnected processes, the system supports automated assessment processing, learner analytics, intervention recommendation, and learning recovery monitoring under the ARAL Program.

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